

Tornadoes

Virtually impossible to predict accurately, tornadoes are the most capricious weather phenomena—and some of the most violent

With a tight vortex of air spinning at up to 300 mph (480 kph), a grade F “twister” is the most terrifying sight in nature, capable of destroying almost anything in its path. To meet the definition of a tornado, the vortex must extend from the base of a cloud and be in contact with the ground.

Types of tornadoes

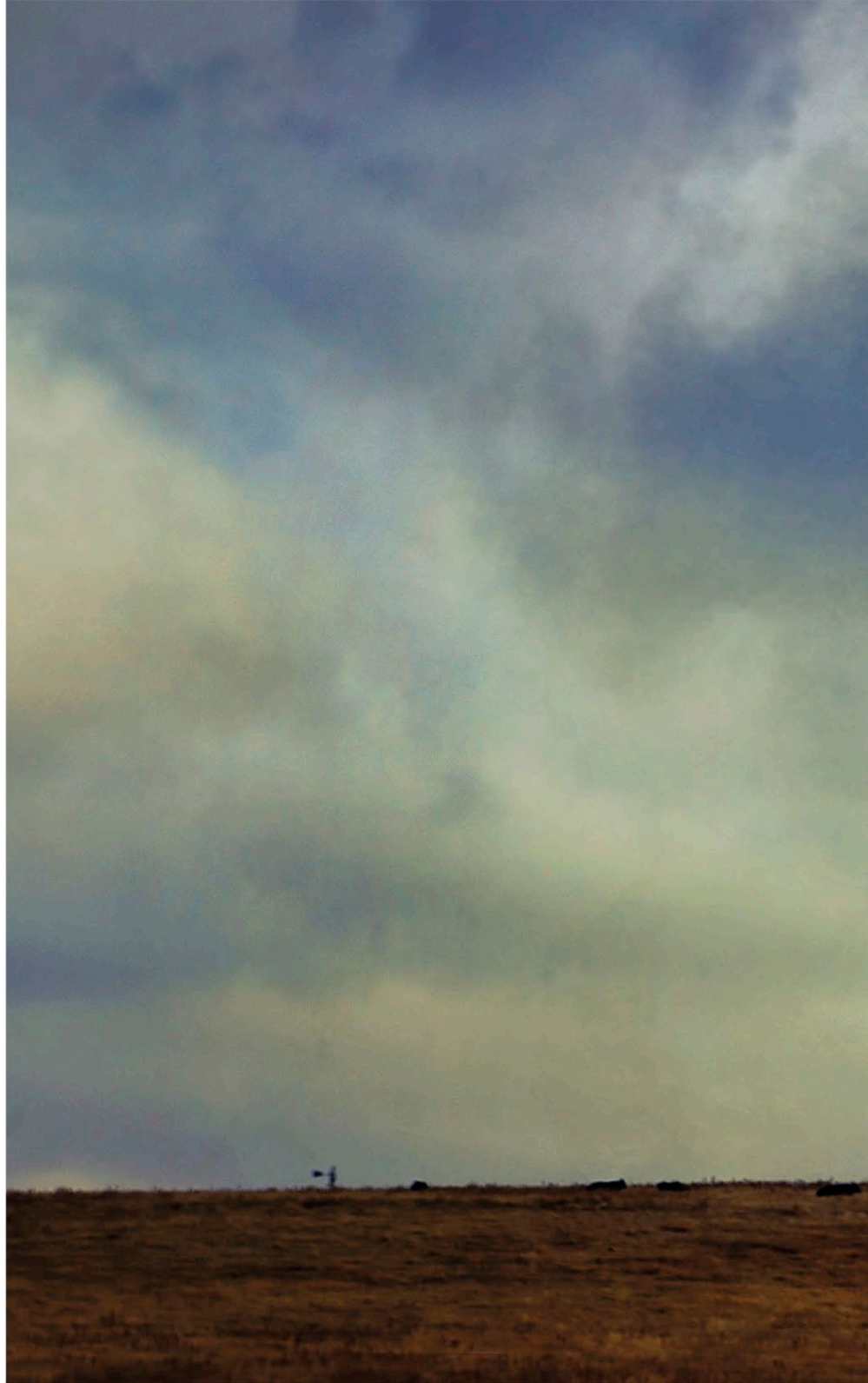
Tornadoes can happen anywhere, but most occur on the Great Plains of the U.S. They occur in different shapes and sizes. Rope tornadoes have a narrow funnel, while wedge tornadoes have a wide diameter at the ground and are even broader at the cloud base. Tornadoes change color depending on the nature of the debris and dust that they pull into their violent vortex and may be near-white, brown, reddish, or nearly black. Their severity is measured on the Fujita scale. An F-0 twister—the most common type—is capable of breaking branches from trees, while an F-5 one is able to carry cars 330 ft (100 m) through the air.

Tornadogenesis

The ideal situation for a tornado to form, a process known as tornadogenesis, is when a cold air mass converges with a warm, moist one, creating instability and towering cumulonimbus clouds. Where there is also wind shear (a change in wind speed or direction with height), a slowly rotating supercell storm may develop (see pp.328–29). If an input of warm, rising air is in contact with the rapidly descending air of the storm’s rear flank downdraft, a narrow column of rapidly spinning air develops. The rotating air continues downward, forming a funnel below the cloud. Air drawn into the funnel enters an area of much lower pressure, so it expands and cools, and moisture condenses as a result. The funnel becomes a tornado on touching the ground.

The biggest twister on record is the Tri-State Tornado of March 18, 1925, which was in contact with the ground for at least 219 miles (352 km). It wreaked havoc across Missouri, Illinois, and Indiana, leaving 695 dead. Bangladesh has the highest annual death toll caused by tornadoes, averaging almost 200 each year.

Bangladesh’s **Daulatpur-Saturia tornado** of April 26, 1989, killed 1,300 people



TORNADO ALLEY

Hundreds of tornadoes are recorded in this region of the central U.S. every year, mostly in spring and early summer. The classic situation for tornadogenesis on the southern Great Plains is a combination of cold, dry air moving southeast from the Rockies and very warm, moist air advancing north from the Gulf of Mexico. This produces atmospheric instability and the growth of large storm cells, some of which produce tornadoes.

